BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

The Capstone black-box optimisation (BBO) exercise involves eight unknown objective functions, with input dimensions ranging from two to eight. Each function is a sealed system, i.e., its analytical form and gradients are unavailable, and the only way to explore it is to submit an input point and observe the output. With a strict limit of one query per function per week, each evaluation was carefully chosen to maximise what was learnt and how effectively the search progressed. I submitted one set of input values every week and receive one number back. The goal is simply to find the highest value each function can return before the attempts run out.

The process is a loop with three parts:
1. First I fit a Gaussian Process to everything observed so far and it returns both a prediction and an honest uncertainty at any untested point.
2. Secondly, an acquisition function weighs predicted value against that uncertainty and proposes where the single available query should go.
3. Thirdly and this is the part that has mattered most - I record what the model predicted alongside what actually came back, and that gap drives the next change rather than the score itself.

A Gaussian Process is the right surrogate here precisely because the second step needs calibrated uncertainty and because it behaves well on smaller datasets. The three parts fit together as a feedback loop:
a. The model proposes
b. The portal disposes
c. The discrepancy between them becomes the evidence for how the next round is conducted.

Across twelve rounds this has produced 263 evaluations spanning two to eight dimensions.

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

Strategy Changes:

In rounds 1-4, I ran the textbook pipeline - fit, maximise an acquisition function over the whole domain, submit whatever it returned.

Rounds 5-7 confined the search to trust regions around the best point, with the exploration weight lowered as evidence accumulated.

From round 8 onwards, I moved one coordinate at a time and put the necessary safeguards into code, like boundary floors, an assertion that exactly one coordinate changes, and a check that each query sits inside a bracket I have already observed.

What influenced them:

In every scenario,the strategy change was driven by Model performance, rather than intuition.
For examples: none of Function 7's four global queries beat the point already present in its starting data. “Function 8’s three multi-coordinate proposals all came back below the incumbent, and none could be diagnosed because seven inputs had moved at once. Moving one coordinate at a time then produced three new bests, and the one query that did not improve the score produced the project’s most informative result. Function 1 forced the largest change - its outputs span more than 100 orders of magnitude, so a Gaussian Process fitted to raw values saw only one reading near −3.6e-3 and was blind to the 1e-16 basin holding the real optimum. Five rounds of careful refinement had been spent on a surface the model could not see. A held-out check exposed it. The log-space replacement predicts within 0.87 to 1.28 times actual, against a miss of roughly 3.8e10 times for the original.
The principles now guiding each query. Keep a method only while it predicts held-out data. Prefer a query whose result is interpretable. Write down what each outcome will mean before the result arrives. And require two methods with different assumptions to agree before reducing a dimension.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

The trend that reshaped my thinking is that spread in the inputs does not indicate influence on the output:
In Function 7 the leading principal component carries 34% of the input variance and loads on x1, x4 and x6, the same three coordinates that clustering had identified independently. But x3 loads only 0.07 on it, the smallest of all six, while being responsible for a measured +0.276, nearly the entire improvement. Keeping only that component would have discarded the one dimension that mattered.
Function 8 shows the reverse trend. Its first component compresses almost nothing, yet correlates −0.878 with the output, recovering the response direction without ever being shown it.

The influential variables are few and often conditional:
Function 3's compound_3 sits pinned between 0.392 and 0.418 across every good result while compounds 1 and 2 move as a single diagonal axis.
Function 5 reduces to a "distance from the corner" axis plus x2 alone.
Function 8's x8 contributes exactly +0.0200 across its range — larger than the entire cumulative gain from three later refinements combined.

How this changed my understanding of the search:
Returns fall away far faster than the data grows: Function 8's successful queries returned +0.2897, then +0.0512, +0.0344 and +0.0023, roughly a 129-fold drop, and Function 7's climb flattened from +0.0428 to +0.00096 per step. More importantly, I now read any summary of the search space with suspicion, because every query since round four was placed by a model optimising against itself. Clustering the result returns a single dense region composed entirely of my own queries. The structure I find is partly the structure I created.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing exploration vs exploitation:
The working rule is to narrow after a good result and widen after a poor one, with the exploration weight treated as evidence-driven rather than scheduled. Function 5's weight ran 2.5 to 0.3 to 0.5 to 0.35 and back to 1.0 as circumstances changed. Late in the project a second rule took over: take the certain gain first because it can be built on, and leave the speculative move last because nothing needs to build on it. Since the score records the best result so far, a failed final query costs the opportunity but never the record.

A decision that worked:
Function 7's surrogate reported x3 as irrelevant for three rounds. The raw data disagreed, so I constrained it by hand against the model's judgement. That produced the project's first real gain and five consecutive records, taking it from 1.364968 to 1.873081, a 37% improvement.

A decision that did not work:
Function 6's acquisition function pushed butter to 0.999604, hard against the domain bound, on smooth extrapolation with no nearby evidence. It returned −0.520103, worse than both the points it was meant to improve on. When the same search pointed back at the same region a round later, I capped the range rather than following it again.

Adapting to surprise:
I treat a large gap between prediction and outcome as a question about the model rather than as noise. Function 8 had frozen x8 for four rounds on the strength of a length-scale reading; one controlled test refuted that at −4.50σ. Every diagnostic had agreed and all were wrong together, because all were downstream of the same data and the same structural assumption.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned actions:
The final query in each function is now pure exploitation, and each has a rule registered in advance. Function 7 converts a bracketed peak that two independent methods locate at the same place (0.5428 by quadratic fit, 0.542 by the model's own argmax). Function 8's remaining candidate scores 0.44 on the gain-to-uncertainty ratio that has so far separated success from failure, below all three of its successes and above both its failures. So if it does not land, the more valuable final move is to repeat the best point and measure whether 9.982957 is reproducible at all, which twelve rounds have never tested. Beyond the budget, I would move to batch proposals, k-fold in place of leave-one-out, and trust-region methods such as TuRBO for the higher-dimensional cases.

Connection to the wider field:
Hyperparameter tuning has exactly this shape. Each configuration costs a full training run, there is no closed form and the budget is finite. This is why Bayesian optimisation is the standard tool there. The same structure governs drug discovery, materials science and process optimisation. The failure mode I hit is general too. A model fitted on data that its behaviour helped select will confirm itself, and possibly internal diagnostic will not be able to identify it.

Explaining it to a non-technical audience:
I had a limited number of attempts to find the best setting of a machine nobody can open. Each result told me where to look next. I wrote down what I expected before every attempt. This enabled me to separate the gains that were real from the ones that were luck. For every decision I made, I have recorded proper justification.